

--	--	--	--	--	--	--	--



B.Tech. Degree III Semester Supplementary Examination April 2019

CS 15-1302 LOGIC DESIGN (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Implement a XOR gate using only NAND gate.
- (b) Convert the following number into new number system
 $(487)_d = (\quad)_b = (\quad)_0 = (\quad)_h$
- (c) Explain the decimal adder circuit.
- (d) Explain the applications of decoders and encoders.
- (e) Write the difference between latches and flip-flops.
- (f) Write a note on ripple counter.
- (g) What is meant by memory decoding?
- (h) Write a note PLA.
- (i) Explain the concept of canonical and standard forms of Boolean equations.
- (j) What is the significance of don't care conditions in Boolean algebra simplification?



PART B

(4 × 10 = 40)

- II. Simplify and implement the following Boolean equation.

$$F = \bar{D}.\bar{C}.B.A + \bar{D}.C.\bar{B}.A + D.\bar{C}.B.A + D.C.\bar{B}.A$$

OR

- III. Design a circuit that receives a 4-bit input representing the integers 0 to 15 and outputs a signal F that is true if the input represents a multiple of 3 or 5 and is false otherwise. Implement the logic circuit using only basic gates.

- IV. Design and implement a Fast Adder circuit in detail.

OR

- V. Design and implement a Binary Adder/subtractor circuit in detail.

- VI. Design and implement a counter which count 0, 4, 8, 12, 15, 2, 6, 0.

OR

- VII. Explain Johnson counters, Ring counters and shift registers in detail.

- VIII. Explain various characteristics of digital logic families.

OR

- IX. Explain basic gate implementations in TTL, RTL, ECL and CMOS logic families.
